



Produce Quality Grading

[One clear sentence explaining what your project does]

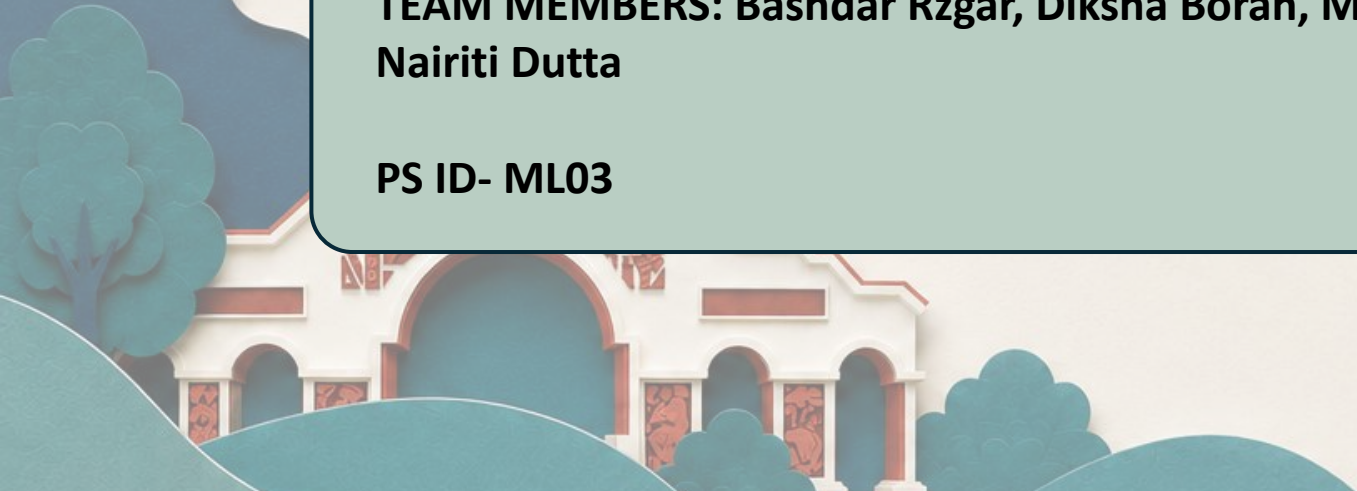


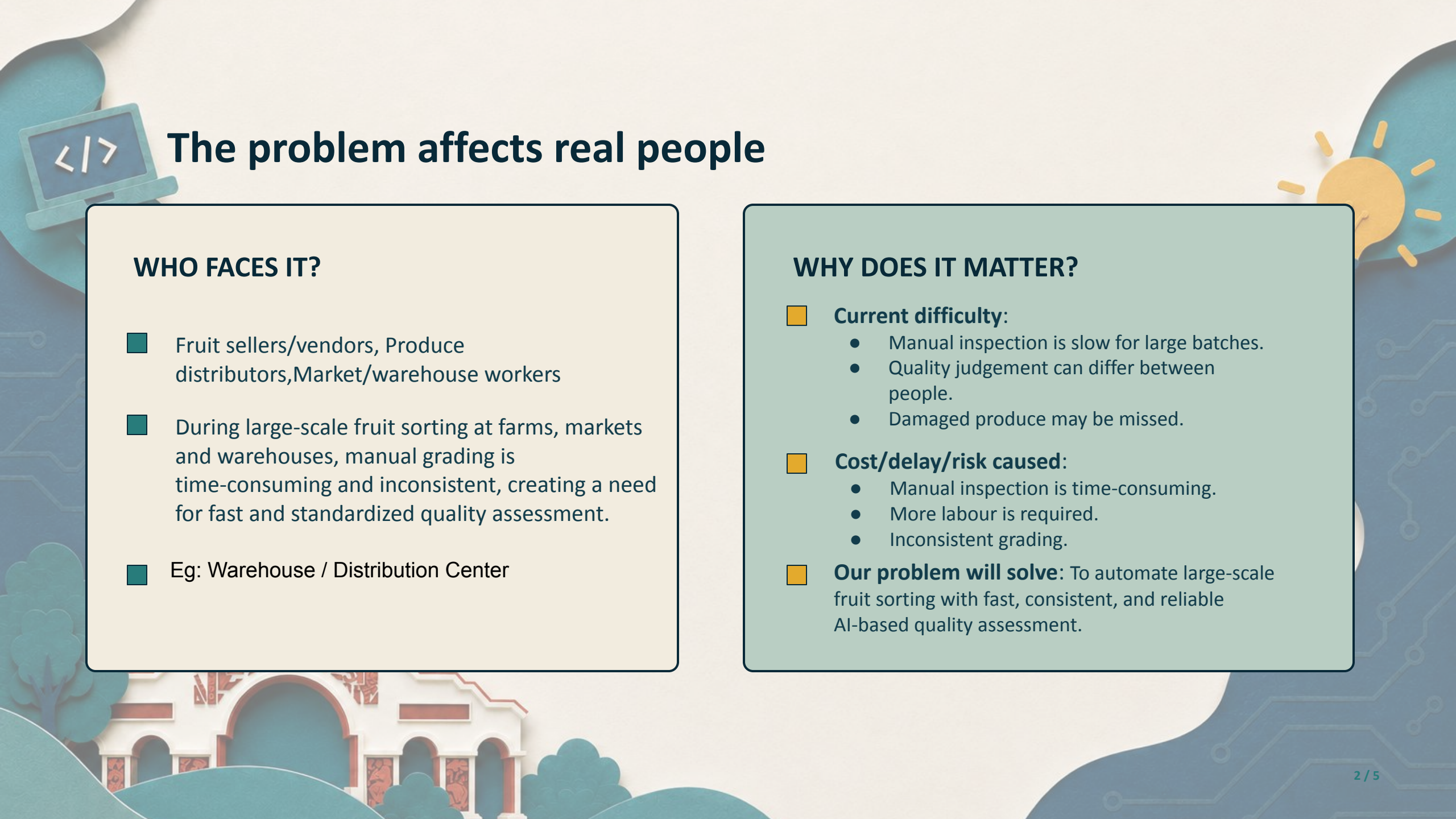
TEAM NAME: CipherCrew

TEAM ID: 65

TEAM MEMBERS: Bashdar Rzgar, Diksha Borah, Manashita Baruah,
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PS ID- ML03





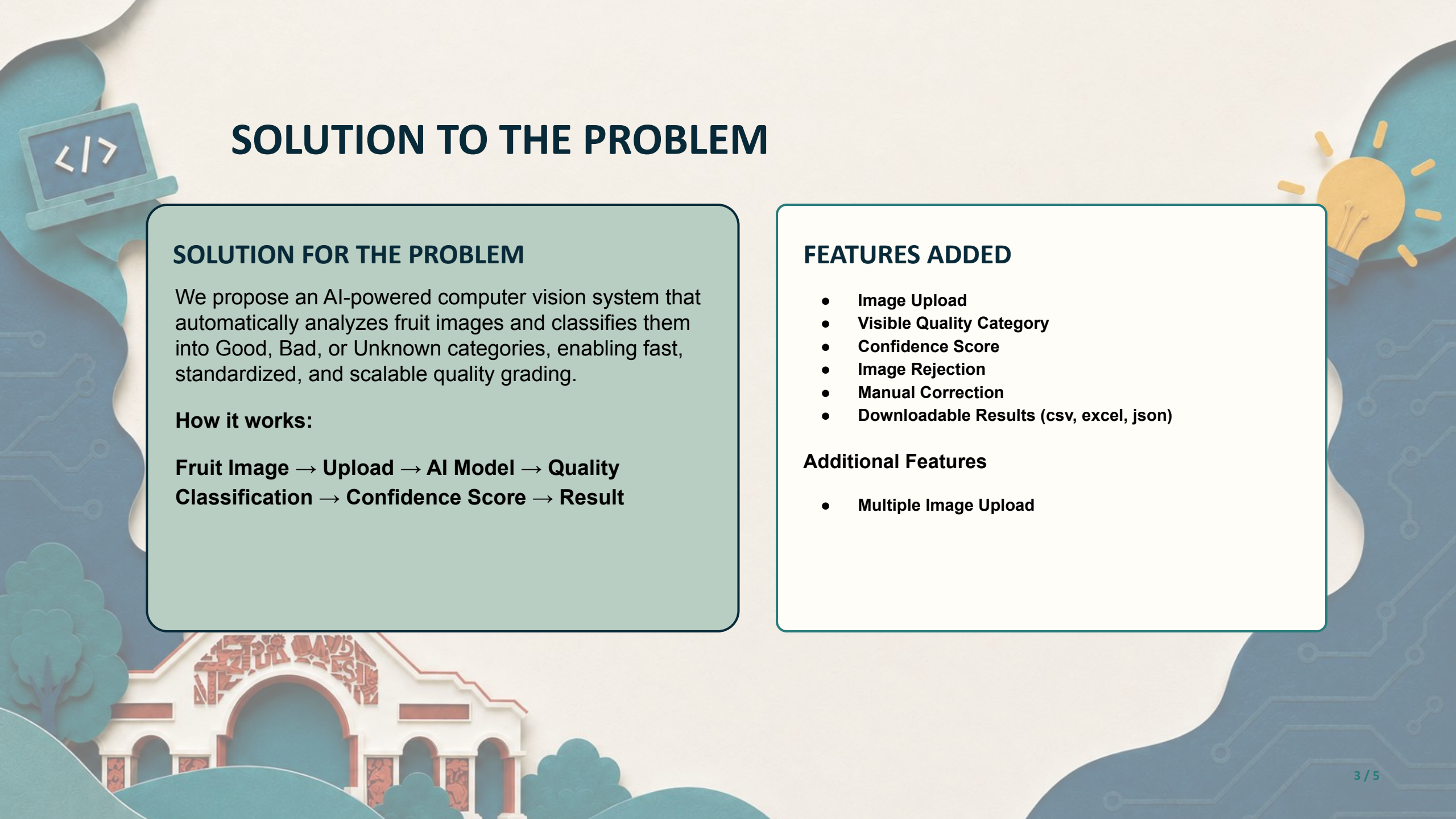
The problem affects real people

WHO FACES IT?

- Fruit sellers/vendors, Produce distributors, Market/warehouse workers
- During large-scale fruit sorting at farms, markets and warehouses, manual grading is time-consuming and inconsistent, creating a need for fast and standardized quality assessment.
- Eg: Warehouse / Distribution Center

WHY DOES IT MATTER?

- **Current difficulty:**
 - Manual inspection is slow for large batches.
 - Quality judgement can differ between people.
 - Damaged produce may be missed.
- **Cost/delay/risk caused:**
 - Manual inspection is time-consuming.
 - More labour is required.
 - Inconsistent grading.
- **Our problem will solve:** To automate large-scale fruit sorting with fast, consistent, and reliable AI-based quality assessment.



SOLUTION TO THE PROBLEM

SOLUTION FOR THE PROBLEM

We propose an AI-powered computer vision system that automatically analyzes fruit images and classifies them into Good, Bad, or Unknown categories, enabling fast, standardized, and scalable quality grading.

How it works:

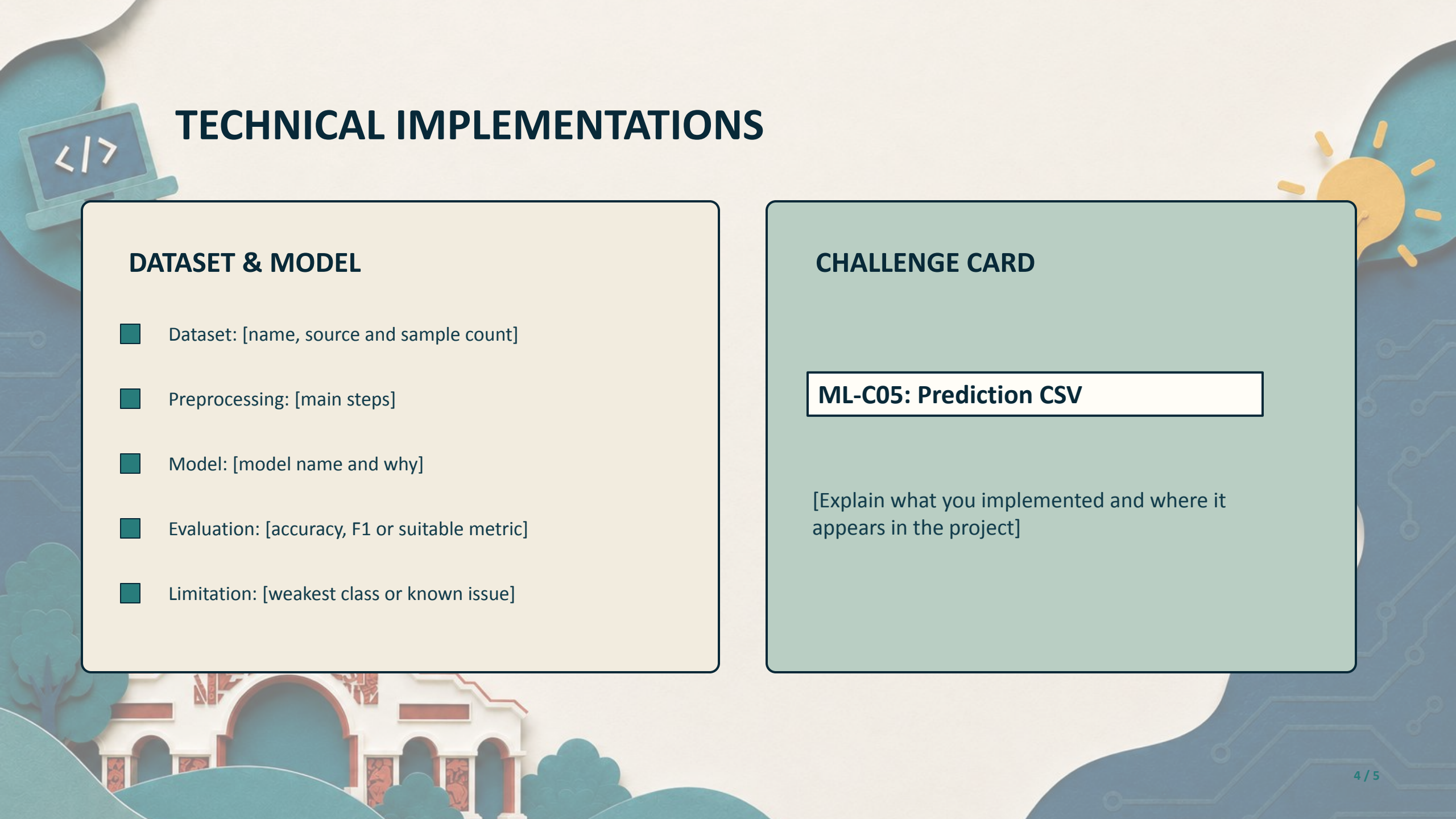
Fruit Image → Upload → AI Model → Quality Classification → Confidence Score → Result

FEATURES ADDED

- Image Upload
- Visible Quality Category
- Confidence Score
- Image Rejection
- Manual Correction
- Downloadable Results (csv, excel, json)

Additional Features

- Multiple Image Upload



TECHNICAL IMPLEMENTATIONS

DATASET & MODEL

- Dataset: [name, source and sample count]
- Preprocessing: [main steps]
- Model: [model name and why]
- Evaluation: [accuracy, F1 or suitable metric]
- Limitation: [weakest class or known issue]

CHALLENGE CARD

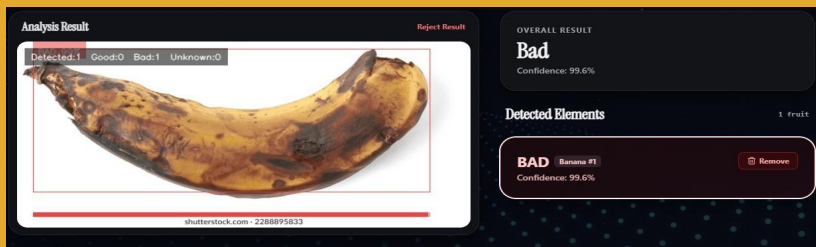
ML-C05: Prediction CSV

[Explain what you implemented and where it appears in the project]

What works today, and what comes next

VALUE

- **Main benefit:** Reduces time-consuming manual quality grading.
- **Who can use it :** Fruit sellers/vendors, Produce distributors, Market/warehouse workers.
- **One measurable or visible result:** Provides an instant quality grade with a confidence score for each produce image.



LIMITATIONS

- **Current technical limit:**
 - Limited dataset & varieties
 - Cannot detect internal damage
 - Single-view images may miss hidden defects
- [Data, privacy or reliability limit]
- **What is not yet included:**
 - Multi-view combined analysis
 - Internal quality detection

THANK YOU